

Precalc Lesson 0.1 Homework

Answer
Key

1. $\sqrt{274} \approx 16.55$

2. $\sqrt{5125} \approx 71.59$ miles

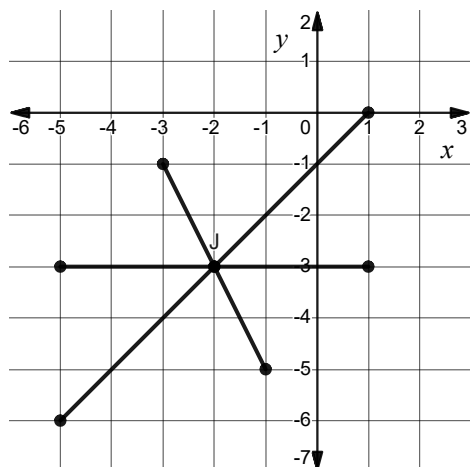
No, most roads run north-south or east-west so they don't interfere with other roads. In Colorado especially, the terrain also has to be taken into account when building a road, so it would be rare that a straight-line path is achievable.

3. $y = 1, y = 13$

4. $a + c = 14$
 $b + d = -20$

5. $(-19, 22)$

6. Many solutions are possible. A sample answer is shown.



7. $j = -1.5, k = 1$

8. Many solutions are possible. The distance between the x-coordinates could be 3 or 4, and the distance between the y-coordinates could be 4 or 3, respectively. Sample solution:
 $A = (1, 5), B = (4, 9)$.
9. All three are correct. Peyton is using the traditional distance formula and Monique is using the Pythagorean theorem which provide the same end result. Maria is considering only positive distances by always subtracting the smaller value from the larger value, which is a valid approach because the horizontal and vertical components represent legs of a right triangle which are always positive. Furthermore, reversing the order of the subtraction does not change the final result since the quantities are squared.